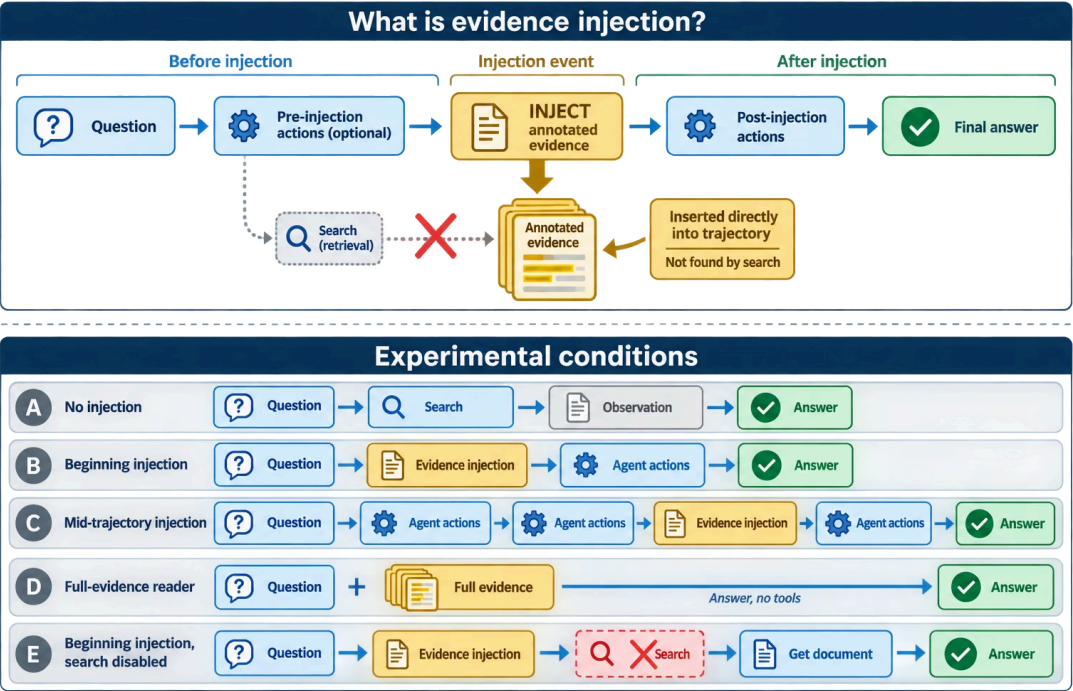


Evidence on a Silver Platter

Finding versus Using Evidence in Agentic Search

I intervene on an agent's trajectory by directly supplying the gold evidence and measure what changes.

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August 4, 2026



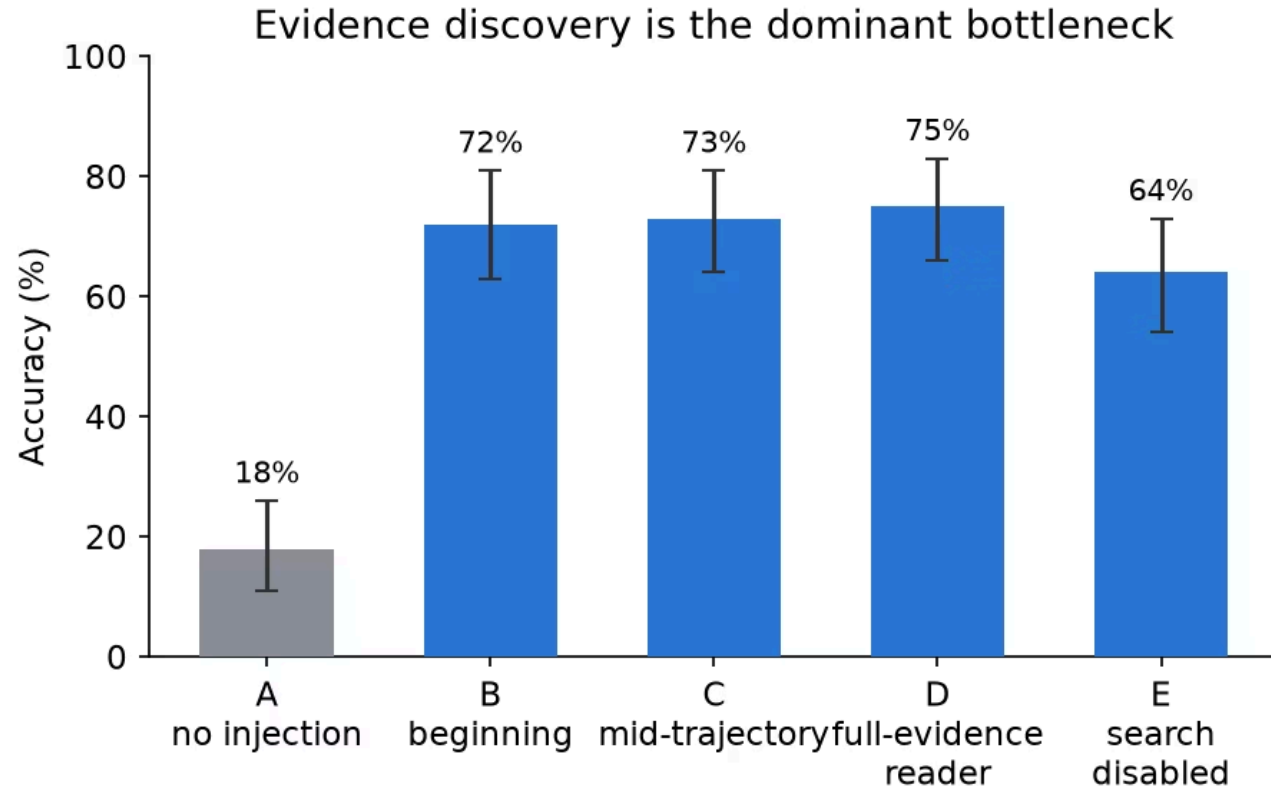
Experimental Design

100 paired questions • BrowseComp-Plus • ReAct agent (GPT-4o)

Fig. 0 – Injection schematic: where the synthetic evidence turn lands in the trajectory.

Condition	Evidence access	Search?
A – no injection	Must be discovered	✓
B – beginning	Injected before action 1	✓
C – mid-trajectory	Injected after action 2	✓
D – full-evidence reader	Long evidence, single prompt	☐
E – search disabled	Injected before action 1	☐

Evidence Discovery is the Dominant Bottleneck



+54 pp

Supplying the evidence raises accuracy by 54 percentage points.

- No injection (A): 18%
- Beginning injection (B): 72%
- Mid-trajectory (C): 73% Full-evidence reader (D): 75% Search disabled (E): 64%

For this agent and question set, most failures arise before the evidence reaches the model.

Fig. 1 – Accuracy across five live conditions ($n = 100$ each; 95% bootstrap CIs).

Timing Changes Efficiency, Not Accuracy

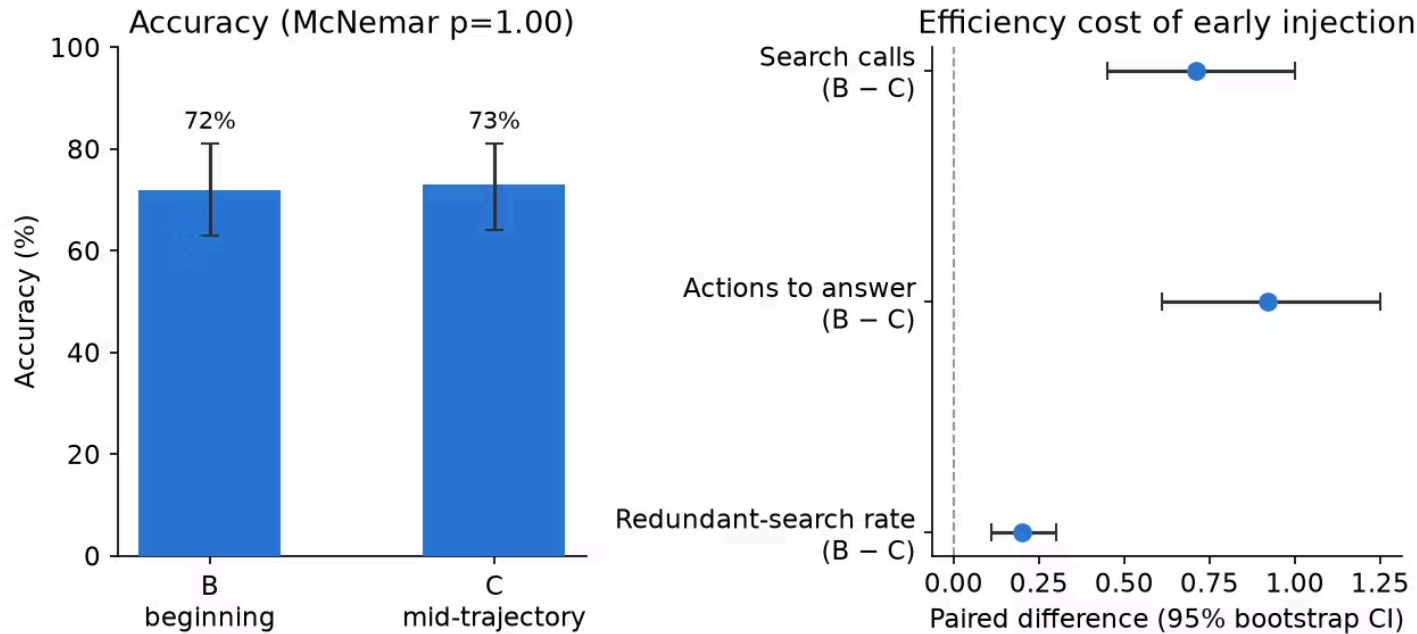


Fig. 2 – Left: B and C are statistically indistinguishable (McNemar $p = 1.00$). Right: beginning injection incurs more post-injection search and actions.

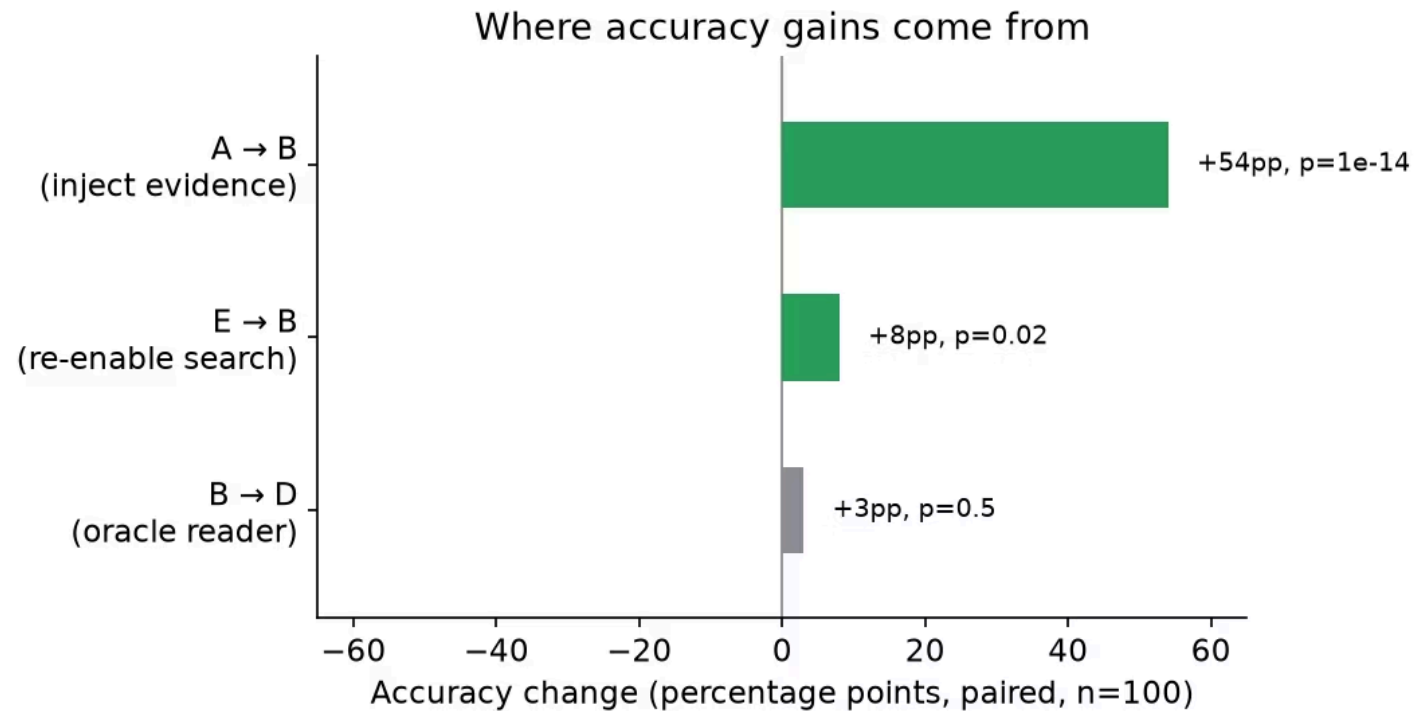
Key findings

- B vs. C accuracy: 72% vs. 73% (McNemar $p = 1.00$)
- Beginning injection: ~0.71 more searches and ~0.92 more actions after injection
- Redundant-search rate ~20 pp higher with beginning injection

Earlier evidence does not improve accuracy, but the agent keeps checking after it already has enough information.

Efficiency differences are paired bootstrap estimates with 95% CIs.

Decomposing the Performance Gap



Three contrasts

- A → B (inject evidence): +54 pp, $p \approx 1 \times 10^{-14}$
- E → B (re-enable search): +8 pp, $p = .02$
- B → D (oracle reader): +3 pp, $p = .51$

*Evidence access explains the largest gain;
additional search provides a smaller benefit;
removing the agentic loop does not produce a
detectable improvement.*

The B–D comparison is not significant and does not establish equivalence.

Fig. 3 – Paired accuracy gains on the same 100 questions ($n = 100$, paired).

Frozen-Trajectory Replay

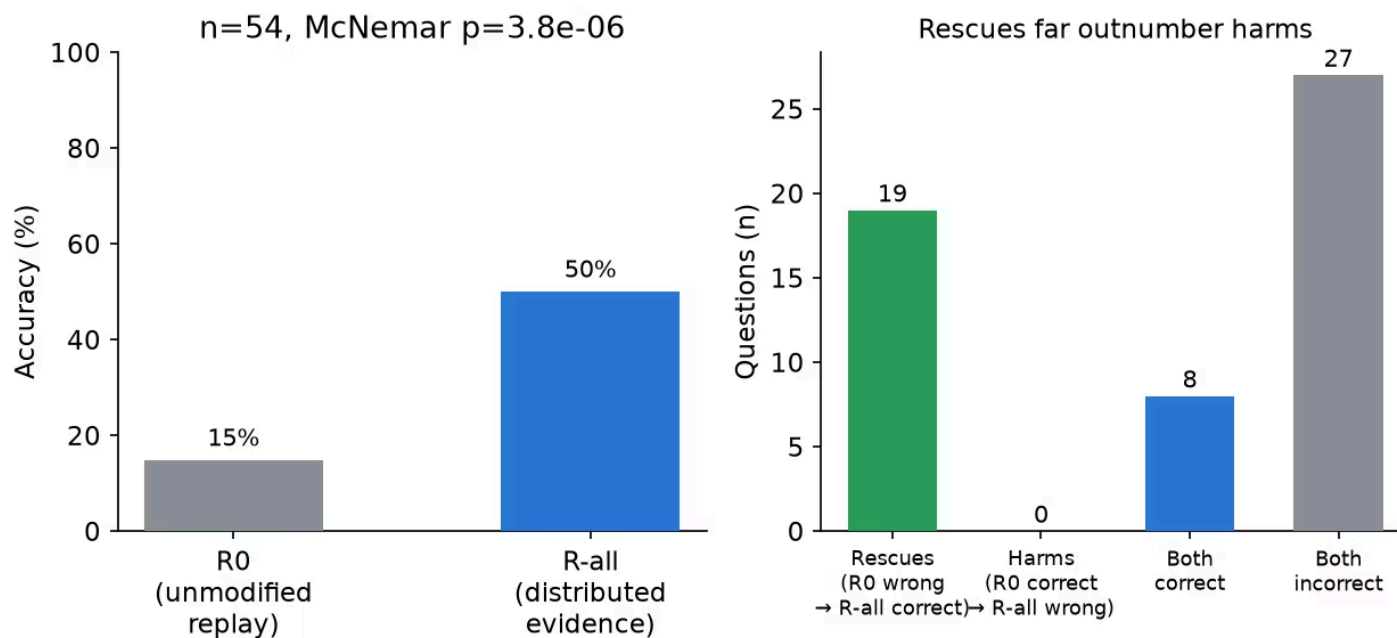


Fig. 4 – Distributed evidence improves frozen-trajectory accuracy. Left: accuracy rises from 15% to 50% ($n = 54$). Right: rescues far outnumber harms.

What was done

Original no-injection trajectories are frozen. Non-evidence documents are replaced with all available gold evidence across the trajectory. Only the final answer is regenerated.

Results ($n = 54$)

- R0 (original replay): 15% → R-all (distributed evidence): 50%
- 19 rescues, 0 harms
- McNemar $p = 3.8 \times 10^{-6}$

Evidence can rescue failed trajectories even without changing the agent's search decisions.

Conclusions and Limitations

Conclusions

- Evidence discovery is the dominant bottleneck in this setting.
- Once gold evidence is available, the agent usually uses it effectively.
- Agents may continue searching even after receiving sufficient evidence.
- Better stopping and evidence-awareness mechanisms could reduce unnecessary search.

Limitations

- One primary agent/model configuration (GPT-4o, ReAct).
- Oracle access to benchmark evidence documents.
- Results from a fixed benchmark and 100-question evaluation.
- Evidence injection does not represent a deployable retrieval system.

The results shift attention from answer generation toward evidence discovery and knowing when to stop.